Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

There are 3 red beads and 5 blue beads. What is the ratio of red to blue? Student's answer: 5:3

Write the ratio in the order asked.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Red comes first in the question, so the ratio is red to blue: 3:5. The student reversed the order.	**3:5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A class has 4 cats and 6 dogs as pets. Write the ratio of cats to dogs in simplest form. Student's answer: 4:6

Simplify the ratio.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Both numbers can be divided by 2. The simplest form is 2:3.	**2:3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

pe uses 2 cups of flour for every 3 cups of sugar. Student says: For every 3 cups of flour, there are 2 cups of

Check the order of the ratio.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The ratio is flour to sugar, 2:3. For every 2 cups of flour there are 3 cups of sugar, not the other way.	**For every 2 cups of flour, there are 3 cups of sugar**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A team won 12 games and lost 8 games. What is the ratio of wins to total games? Student's answer: 12:8

Find the total number of games.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The total is 12 + 8 = 20 games. Wins to total is 12:20, which simplifies to 3:5.	**3:5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The ratio of apples to oranges is 1:3. Student says: The ratio of 15 apples to 5 oranges is also 1:3.

Compare the two ratios.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The numbers were switched. 15 apples to 5 oranges is 15:5, which simplifies to 3:1.	**3:1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the ratio 6:9 equivalent to 2:3? Student says: No

Simplify or scale the ratio.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Divide both numbers in 6:9 by 3 to get 2:3. The ratios are equivalent.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A car drives 180 miles in 3 hours. What is its unit rate in miles per hour? Student's answer: 3

Divide to find the unit rate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided the time by the distance. Divide the distance by the time: 180 ÷ 3 = 60 miles per hour.	**60 miles per hour**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Five pounds of apples cost $15. What is the price per pound? Student's answer: $10

Divide the cost by the pounds.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 5 from 15. Divide: $15 ÷ 5 = $3 per pound.	**$3 per pound**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is the better buy: 8 cans for $6 or 5 cans for $4? Student says: 5 cans for $4

Find the price per can for each.

What did the student do wrong? Why do you think the student made this error?

8 cans for $6 is $0.75 per can. 5 cans for $4 is $0.80 per can. The lower unit price is the better buy.

Solve the problem correctly. Show all of your work.

8 cans for $6

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A car uses 12 gallons of gas to drive 300 miles. How many miles per gallon? Student's answer: 36

Divide to find the unit rate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 12 and 300 and then made a mistake. Divide 300 by 12 to get 25 miles per gallon.	**25 miles per gallon**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A runner goes 4 miles in 30 minutes. How many minutes per mile? Student's answer: 8 minutes

Divide to find minutes per mile.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student rounded 7.5 up to 8. The exact answer is 30 ÷ 4 = 7.5 minutes per mile.	**7.5 minutes per mile**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3 bananas cost $1.20. What is the cost of 1 banana? Student's answer: $3.60

Divide to find the cost of one.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied by 3. Divide: $1.20 ÷ 3 = $0.40.	**$0.40**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3 bags cost $9 and 5 bags cost $15. What is the cost of 10 bags? Student's answer: $25

Use the table or a unit rate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each bag costs $3. Ten bags cost 10 × $3 = $30. The student added 10 to 15.	**$30**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is 25% of 60? Student's answer: 6

Find the percent of the number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found 10% instead of 25%. 25% of 60 is 0.25 × 60 = 15.	**15**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

On a map, 1 inch equals 40 miles. How many miles are 2.5 inches? Student's answer: 80 miles

Use the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 2 inches instead of 2.5. 2.5 × 40 = 100 miles.	**100 miles**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sweater costs $40 and is 30% off. What is the sale price? Student's answer: $12

Find the discount, then the sale price.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$12 is the amount of the discount. The sale price is $40 - $12 = $28.	**$28**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ratio of boys to girls in a class is 3:2. There are 30 students total. How many girls are there? Student's answe

Use the ratio parts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found the number of boys. There are 5 parts total, so each part is 6 students. Girls are 2 parts: 2 × 6 = 12.	**12 girls**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A student answers 18 out of 24 quiz questions correctly. What percent is correct? Student's answer: 18%

Write the fraction and change it to a percent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the number correct, not the percent. 18/24 = 3/4 = 75%.	**75%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 2/3 ÷ 1/4. Student's answer: 1/6

Multiply by the reciprocal.

What did the student do wrong? Why do you think the student made this error?

The student multiplied the fractions. Multiply by the reciprocal of 1/4, which is 4: 2/3 × 4 = 8/3 = 2 2/3.

Solve the problem correctly. Show all of your work.

8/3 or 2 2/3

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 3/4 ÷ 3. Student's answer: 9/4

Multiply by the reciprocal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied by 3 instead of dividing. 3 is 3/1, so its reciprocal is 1/3. 3/4 × 1/3 = 1/4.	**1/4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 5/6 ÷ 2/3. Student's answer: 5/9

Multiply by the reciprocal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided the numerators and the denominators. Multiply by 3/2: 5/6 × 3/2 = 15/12 = 5/4 = 1 1/4.	**5/4 or 1 1/4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many 1/4-cup servings are in 3 1/2 cups? Student's answer: 3 1/8

Divide the total by the serving size.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied. Divide: 3 1/2 ÷ 1/4 = 7/2 × 4 = 14 servings.	**14 servings**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 7 ÷ 1/2. Student's answer: 7/2

Multiply by the reciprocal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided 7 by 2 instead of using the reciprocal. 7 ÷ 1/2 = 7 × 2 = 14.	**14**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 2 1/4 ÷ 3/4. Student's answer: 3/4

Change to improper fractions, then multiply by the reciprocal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided 9/4 by 3 instead of by 3/4. 9/4 × 4/3 = 36/12 = 3.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 4.8 ÷ 0.6. Student's answer: 0.8

Move the decimal point in both numbers.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Move the decimal one place in each number to get 48 ÷ 6 = 8.	**8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 12.6 ÷ 0.7. Student's answer: 1.8

Move the decimal point in both numbers.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student moved only one decimal point. Use 126 ÷ 7 = 18.	**18**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 36.75 ÷ 2.5. Student's answer: 1.47

Move the decimal point in both numbers, then divide.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student moved the decimal point the wrong way. Use 367.5 ÷ 25 = 14.7.	**14.7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 8.4 ÷ 0.4. Student's answer: 2.1

Move the decimal point in both numbers.

What did the student do wrong? Why do you think the student made this error?

The student moved only the decimal in the divisor. Use 84 ÷ 4 = 21.

Solve the problem correctly. Show all of your work.

21

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 0.2 ÷ 0.05. Student's answer: 0.4

Move the decimal point in both numbers.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Move the decimal two places in each number: 20 ÷ 5 = 4.	**4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A ribbon is 3.6 meters long. How many 0.4-meter pieces can be cut from it? Student's answer: 0.9

Divide the length by the piece length.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided the wrong way. 3.6 ÷ 0.4 = 36 ÷ 4 = 9 pieces.	**9 pieces**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 3.25 + 4.8. Student's answer: 7.13

Line up the decimal points.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not line up the decimals. Write 4.80 and add: 3.25 + 4.80 = 8.05.	**8.05**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract 9.4 - 2.67. Student's answer: 7.73

Line up the decimal points and subtract.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Write 9.40 and subtract. 9.40 - 2.67 = 6.73. The student made a subtraction error in the hundredths place.	**6.73**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 2.5 × 4.6. Student's answer: 115

Multiply as whole numbers, then place the decimal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Two decimal places total, but the student did not place the decimal. 25 × 46 = 1,150, so 2.5 × 4.6 = 11.5.	**11.5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 0.75 × 0.4. Student's answer: 0.030

Multiply as whole numbers, then place the decimal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
75 × 4 = 300, and there are 3 decimal places total, so the answer is 0.300, which is 0.3.	**0.3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 6.84 + 3.16. Student's answer: 9.9

Line up the decimal points and add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The hundredths add to 10, so carry. 6.84 + 3.16 = 10.00.	**10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Four items cost $3.45 each. What is the total cost? Student's answer: $12.45

Multiply the price by 4.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 3.45 + 9. Multiply: 4 × $3.45 = $13.80.	**$13.80**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the greatest common factor of 24 and 36. Student's answer: 6

List the factors of each number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
6 is a common factor, but it is not the greatest. The factors of 24 and 36 include 12, which is the greatest.	**12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the least common multiple of 6 and 8. Student's answer: 48

List the multiples of each number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
48 is a common multiple, but 24 is smaller. The least common multiple is 24.	**24**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Use the distributive property: 4(3 + 5). Student's answer: 17

Multiply 4 by each term inside the parentheses.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 4 × 3 and then added 5. Multiply 4 by both terms: 4 × 3 + 4 × 5 = 12 + 20 = 32.	**32**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 18 + 12 as a product using the greatest common factor. Student writes: 3(6 + 4)

Factor out the greatest common factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 is a factor, but it is not the greatest. The GCF of 18 and 12 is 6, so the answer is 6(3 + 2).	**6(3 + 2)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the greatest common factor of 14 and 35. Student's answer: 1

List the factors of each number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
7 is a factor of both 14 and 35, so it is the greatest common factor.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

d 36 blue beads. She wants to make identical bags with the most bags possible, using all the beads. How n

Find the greatest common factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
6 bags would work, but the greatest number of bags is the GCF of 48 and 36, which is 12.	**12 bags**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The temperature is -5°C. It rises 8 degrees. What is the new temperature? Student's answer: -13°C

Start at -5 and move up 8.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Rising means moving up on the number line. -5 + 8 = 3°C.	**3°C**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bank account has $50. A withdrawal of $80 is made. What is the new balance? Student's answer: $30

Use a negative number for money owed.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The account is overdrawn, so the balance is negative. $50 - $80 = -$30.	**-$30**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is less: -3 or -7? Student's answer: -3

Compare the numbers on a number line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
-7 is farther to the left on the number line, so it is less than -3.	**-7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A diver is 40 feet below sea level. Write this as a signed number. Student's answer: 40 feet

Use a negative number for below sea level.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Below sea level is a negative location. The answer is -40 feet.	**-40 feet**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 0 a positive number or a negative number? Student's answer: positive

Think about where zero sits on the number line.

What did the student do wrong? Why do you think the student made this error?

Zero is neither positive nor negative. It is the number between positive and negative numbers.

Solve the problem correctly. Show all of your work.

Neither

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the opposite of -12? Student's answer: -12

Find the number the same distance from 0 on the other side.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The opposite of -12 is the same distance from zero on the other side, which is 12.	**12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the opposite of 7/2? Student's answer: 2/7

Change only the sign.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student flipped the fraction. The opposite of 7/2 is -7/2.	**-7/2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In which quadrant is the point (-3, 4)? Student's answer: first quadrant

Check the sign of each coordinate.

What did the student do wrong? Why do you think the student made this error?

The x-coordinate is negative and the y-coordinate is positive. That is the second quadrant.

Solve the problem correctly. Show all of your work.

Second quadrant

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The point (2, -5) is reflected across the y-axis. What is the new point? Student's answer: (2, -5)

Change the sign of the x-coordinate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Reflecting across the y-axis changes the sign of x. The new point is (-2, -5).	**(-2, -5)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Where is the point (0, -6)? Student says: on the x-axis

Look at the coordinates.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The x-coordinate is 0, so the point is on the y-axis, 6 units below the origin.	**On the y-axis**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

On a number line, which is greater: -1.5 or -0.5? Student's answer: -1.5

Compare the positions on the number line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
-0.5 is closer to zero, so it is greater than -1.5.	**-0.5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In which quadrant is the point (-4, -1)? Student's answer: first quadrant

Check the sign of each coordinate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Both coordinates are negative. That is the third quadrant.	**Third quadrant**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the absolute value of -9? Student's answer: -9

Find the distance from zero.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Absolute value is distance from zero, which is never negative. |-9| = 9.	**9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Order these numbers from least to greatest: -2, 1, -5, 0. Student's answer: -2, -5, 0, 1

Place the numbers on a number line.

What did the student do wrong? Why do you think the student made this error?

The student ordered the negatives wrong. -5 is less than -2, so the order is -5, -2, 0, 1.

Solve the problem correctly. Show all of your work.

-5, -2, 0, 1

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is -4 greater than |-6|? Student says: Yes

Find the absolute value first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
|-6| = 6, and 6 is greater than any negative number. So -4 is not greater.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Temperatures are -12°F and -3°F. Which is colder? Student's answer: -3°F

Compare the numbers on a number line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
-12 is farther below zero, so it is colder.	**-12°F**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is less: a debt of -$45 or a debt of -$30? Student's answer: -$30

Compare the numbers on a number line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
-45 is farther left on the number line, so it is less.	**-$45**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is |-15| + |4|? Student's answer: -11

Find each absolute value first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Absolute values are never negative. |-15| = 15 and |4| = 4, so the sum is 19.	**19**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance from (0, 0) to (0, 7). Student's answer: 0

Count the units between the points.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The points are 7 units apart along the y-axis. The distance is 7.	**7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (2, 3) and (8, 3). Student's answer: 5

Subtract the x-coordinates.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student miscounted. 8 - 2 = 6, so the distance is 6 units.	**6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (-4, 1) and (3, 1). Student's answer: 4

Find the distance on the same horizontal line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only from -4 to 0. Use 3 - (-4) = 7 units.	**7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (-2, 5) and (-2, -3). Student's answer: 2

Find the distance on the same vertical line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted wrong. 5 - (-3) = 8 units.	**8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A point is at (-6, 0). What is its distance from the origin? Student's answer: -6

Find the distance from (0, 0).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Distance is never negative. The point is 6 units from the origin.	**6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

d a park is at (1, 2) on a city grid where each unit is one block. How many blocks is the school from the park

Subtract the y-coordinates.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 9 and 2. Subtract: 9 - 2 = 7 blocks.	**7 blocks**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 4^3. Student's answer: 12

Write the power as repeated multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 4×3. $4^3 = 4 \times 4 \times 4 = 64$.	**64**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 2. Student's answer: 10

Write the power as repeated multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 2 × 5. 2 = 2 × 2 × 2 × 2 × 2 = 32.	**32**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate $3 + 2^2$. Student's answer: 25

Apply the exponent before adding.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student squared 5 instead of 2. Only 2 is raised to the power: $2^2 = 4$, so $3 + 4 = 7$.	**7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 10^3. Student's answer: 300

Write the power as repeated multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 10×3. $10^3 = 10 \times 10 \times 10 = 1{,}000$.	**1,000**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 7 × 7 × 7 × 7 using an exponent. Student writes: 4

Write the base and the count.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The base is 7, and it is used 4 times. The answer is 7.	**7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 0.3^2. Student's answer: 0.6

Write the power as repeated multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 0.3 by 2. $0.3^2 = 0.3 \times 0.3 = 0.09$.	**0.09**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an expression for: 5 less than 3 times n. Student writes: 5 - 3n

Write the expression in the order the words say.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 times n is 3n, and 5 less than that is 3n - 5. The order of the words matters.	**3n – 5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 2a + 3b when a = 4 and b = 5. Student's answer: 14

Substitute and multiply first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added all the numbers. Substitute, then multiply: 2 × 4 + 3 × 5 = 8 + 15 = 23.	**23**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate $x^2 - 4$ when $x = 5$. Student's answer: 1

Square the number first, then subtract.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted before squaring. $x^2 = 25$, so $25 - 4 = 21$.	**21**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an expression for: a number n plus 7, then doubled. Student writes: 2n + 7

Use parentheses to show the doubling.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Doubling applies to the whole sum. The expression is 2(n + 7).	**2(n + 7)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate $3 \times (y + 2)$ when $y = 1$. Student's answer: 5

Substitute, then work in the parentheses.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not multiply by 3. $3 \times (1 + 2) = 3 \times 3 = 9$.	**9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an expression for: the product of 4 and a number k, minus 9. Student writes: 9 - 4k

Write the expression in the order the words say.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The product is 4k, and 9 is subtracted from it. The expression is 4k - 9.	**4k – 9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Expand 3(x + 4). Student writes: 3x + 4

Multiply 3 by each term.

What did the student do wrong? Why do you think the student made this error?

3 must multiply both terms inside the parentheses. The answer is 3x + 12.

Solve the problem correctly. Show all of your work.

3x + 12

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify 5x + 2x + 7. Student writes: 10x + 7

Combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the coefficients wrong. 5x + 2x = 7x, so the answer is 7x + 7.	**7x + 7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor 12 + 8x using the greatest common factor. Student writes: 2(6 + 4x)

Find the greatest common factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2 is a factor, but it is not the greatest. The GCF of 12 and 8x is 4, so the answer is 4(3 + 2x).	**4(3 + 2x)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Expand 2(a - 5). Student writes: 2a - 5

Multiply 2 by each term.

What did the student do wrong? Why do you think the student made this error?

The student did not multiply the 5 by 2. The answer is 2a - 10.

Solve the problem correctly. Show all of your work.

2a - 10

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify 4y - y + 3. Student writes: 5y + 3

Combine like terms.

What did the student do wrong? Why do you think the student made this error?

The student added 4y and y. Subtract: 4y - y = 3y, so the answer is 3y + 3.

Solve the problem correctly. Show all of your work.

3y + 3

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Expand $-2(x + 3)$. Student writes: $-2x + 6$

Multiply carefully, including the negative sign.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a sign error on the 3. $-2 \times 3 = -6$, so the answer is $-2x - 6$.	**$-2x - 6$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are $2(x + 3)$ and $2x + 3$ equivalent? Student says: Yes

Expand the first expression and compare.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2(x + 3) = 2x + 6$, which is not $2x + 3$. The expressions are not equivalent.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are $4(x + 2)$ and $4x + 8$ equivalent? Student says: No

Expand the first expression and compare.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$4(x + 2) = 4x + 8$. The expressions are equivalent.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which expression is equivalent to 3(2x + 3)? Student says: 6x + 3

Multiply 3 by each term.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 × 3 = 9, not 3. The equivalent expression is 6x + 9.	**6x + 9**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is x + x + x equal to 3x? Student says: No

Combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Three copies of x make 3x. The expressions are equal.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are 5a + 2 and 2 + 5a equivalent? Student says: No

Think about the order of addition.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Addition can be done in any order, so the expressions are equivalent.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $2m + 3m$ equal to $5m^2$? Student says: Yes

Combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2m + 3m = 5m$. Adding does not square the variable, so $5m^2$ is not equal.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is x = 4 a solution of x + 7 = 11? Student says: No

Substitute 4 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
4 + 7 = 11, which is true. So x = 4 is a solution.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is x = 3 a solution of 2x > 8? Student says: Yes

Substitute 3 for x and check.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2 × 3 = 6, and 6 is not greater than 8. So x = 3 is not a solution.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $x = 5$ a solution of $3x < 12$? Student says: Yes

Substitute 5 for x and check.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$3 \times 5 = 15$, and 15 is not less than 12. So $x = 5$ is not a solution.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is y = 0 a solution of 4y = 10? Student says: Yes

Substitute 0 for y and check.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
4 × 0 = 0, which is not 10. So y = 0 is not a solution.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is n = 6 a solution of n + 4 ≥ 10? Student says: No

Substitute 6 for n and check.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
6 + 4 = 10, and 10 is greater than or equal to 10. So n = 6 is a solution.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is p = -2 a solution of 2p + 7 = 3? Student says: No

Substitute -2 for p and check.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2(-2) + 7 = -4 + 7 = 3, which is true. So p = -2 is a solution.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Tickets cost $8 each. Write an expression for the cost of n tickets. Student writes: n + 8

Decide whether to add or multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each ticket costs $8, so the cost of n tickets is found by multiplying. The expression is 8n.	**8n**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Sam is 4 years older than Ann. Let a be Ann's age. Write an expression for Sam's age. Student writes: a - 4

Decide whether older means more or less.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Sam is older, so his age is more than Ann's. The expression is a + 4.	**a + 4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ben has x marbles and gives away 5. Write an expression for how many he has left. Student writes: 5 - x

Write the expression in the order of the story.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Ben starts with x and gives away 5, so the remaining amount is x - 5.	**x – 5**

Original task. Format inspired by error analysis routines.

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Find the Error

6.EE.6

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pens cost $2 each plus a $3 fee. Write an expression for the cost of p pens. Student writes: 2(p + 3)

Decide which part is multiplied.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Only the pens are charged $2 each. The fee is added once, so the expression is 2p + 3.	**2p + 3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ctangle has width w. The length is 2 more than the width. Write an expression for the area. Student writes: w

Multiply the length by the width.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Area is length times width. The length is w + 2, so the area is w(w + 2).	**w(w + 2)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an expression for the number of hours in d days. Student writes: d + 24

Decide whether to add or multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each day has 24 hours, so the total is found by multiplying. The expression is 24d.	**24d**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve x + 9 = 14. Student's answer: 23

Use the inverse operation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 9 to both sides. Subtract 9 from both sides: x = 5.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve 4x = 36. Student's answer: 32

Use the inverse operation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 4 instead of dividing. Divide both sides by 4: x = 9.	**9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x \div 5 = 7$. Student's answer: 12

Use the inverse operation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 5 to 7. Multiply both sides by 5: x = 35.	**35**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve 3.5 + x = 8. Student's answer: 11.5

Use the inverse operation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 3.5 to 8. Subtract 3.5 from both sides: x = 4.5.	**4.5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve -6 + x = 2. Student's answer: -4

Use the inverse operation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 6 from 2. Add 6 to both sides: x = 8.	**8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $(2/3)x = 4$. Student's answer: 8/3

Multiply both sides by the reciprocal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied by 2/3 instead of 3/2. Multiply both sides by 3/2: x = 6.	**6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an inequality for: x is greater than 5. Student writes: x < 5

Match the symbol to the words.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Greater than means the symbol points to the smaller number. The inequality is x > 5.	**x > 5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an inequality for: at most 12 students. Student writes: $n < 12$

Think about whether 12 is included.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
At most includes 12. The inequality is $n \leq 12$.	**$n \leq 12$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x - 4 > 3$. Student's answer: $x > -1$

Add 4 to both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 4. Add 4 to both sides: $x > 7$.	**$x > 7$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3y \leq 15$. Student's answer: $y \geq 5$

Divide both sides by 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Dividing by a positive number does not change the direction. The answer is $y \leq 5$.	**$y \leq 5$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is −3 a solution of $x \geq -2$? Student says: Yes

Check whether −3 is in the solution set.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
−3 is less than −2, so it does not satisfy $x \geq -2$.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an inequality for: x is less than or equal to 0. Student writes: $x < 0$

Think about whether 0 is included.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Less than or equal to includes 0. The inequality is $x \leq 0$.	**$x \leq 0$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The cost C is 5 times the hours t. Which variable is independent? Student's answer: C

Find the variable that is chosen first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The cost depends on the hours. Hours is the independent variable, so the answer is t.	**t**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A car travels d = 60h miles in h hours. Which is the dependent variable? Student's answer: h

Find the variable that depends on the other.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The distance depends on the hours, so d is the dependent variable.	**d**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A plant's height is h = 2 + 3w inches after w weeks. Which is the independent variable? Student's answer: h

Find the variable that is chosen first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The height depends on the number of weeks. Weeks is the independent variable, so the answer is w.	**w**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Earnings E = 7 × hours worked. Which is the independent variable? Student's answer: earnings

Find the variable that is chosen first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Earnings depend on hours worked. Hours is the independent variable.	**Hours worked**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Savings S = 50 + 10m, where m is the number of months. Is S independent? Student says: Yes

Check whether S depends on m.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
S depends on m, so S is the dependent variable and is not independent.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

For y = x + 4, what is y when x = 3? Student's answer: 12

Substitute 3 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 3 by 4. Add: 3 + 4 = 7.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has a base of 10 and a height of 6. What is its area? Student's answer: 60

Use the triangle area formula.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot to multiply by 1/2. Area = 1/2 × 10 × 6 = 30.	**30**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A parallelogram has a base of 8 and a height of 5. What is its area? Student's answer: 26

Use the parallelogram area formula.

What did the student do wrong? Why do you think the student made this error?

The student added the sides. Area = base × height = 8 × 5 = 40.

Solve the problem correctly. Show all of your work.

40

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A trapezoid has bases of 6 and 10 and a height of 4. What is its area? Student's answer: 64

Add the bases, then multiply by height and divide by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot to divide by 2. Area = (6 + 10) × 4 ÷ 2 = 32.	**32**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has a base of 7 and a height of 4. What is its area? Student's answer: 28

Use the triangle area formula.

What did the student do wrong? Why do you think the student made this error?

The student forgot to multiply by 1/2. Area = 1/2 × 7 × 4 = 14.

Solve the problem correctly. Show all of your work.

14

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rhombus has diagonals of 6 and 8. What is its area? Student's answer: 48

Use the rhombus area formula.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot to divide by 2. Area = 6 × 8 ÷ 2 = 24.	**24**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has a base of 3 1/2 and a height of 4. What is its area? Student's answer: 14

Use the triangle area formula.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot the 1/2 in the formula. Area = 1/2 × 3.5 × 4 = 7.	**7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a box that is 1/2 unit long, 3 units wide, and 2 units tall. Student's answer: 5.5 cubic units

Multiply the three edges.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the edges. Multiply: 1/2 × 3 × 2 = 3 cubic units.	**3 cubic units**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ind the volume of a box that is 2 1/2 units long, 2 units wide, and 3 units tall. Student's answer: 7.5 cubic unit

Multiply the three edges.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student left out the width. 2.5 × 2 × 3 = 15 cubic units.	**15 cubic units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a cube with edges of 1/2 foot. Student's answer: 1/4 cubic foot

Multiply all three edges.

What did the student do wrong? Why do you think the student made this error?

The student multiplied only two edges.
(1/2) × (1/2) × (1/2) = 1/8 cubic foot.

Solve the problem correctly. Show all of your work.

1/8 cubic foot

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a box that is 3/4 unit long, 2 units wide, and 4 units tall. Student's answer: 4 cubic units

Multiply the three edges.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student left out the length. 3/4 × 2 × 4 = 6 cubic units.	**6 cubic units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 4 inches by 2 inches by 1 inch. How many 1/2-inch cubes fill it? Student's answer: 16

Find the volume of the box and the volume of one cube.

What did the student do wrong? Why do you think the student made this error?

The box has a volume of 8 cubic inches. Each small cube has a volume of 1/8 cubic inch. 8 ÷ 1/8 = 64 cubes.

Solve the problem correctly. Show all of your work.

64 cubes

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a box that is 1 1/2 units long, 1 unit wide, and 2/3 unit tall. Student's answer: 2 cubic units

Multiply the three edges.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a multiplication error. 3/2 × 1 × 2/3 = 1 cubic unit.	**1 cubic unit**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A square has vertices at (0, 0), (4, 0), (4, 4), and (0, 4). What is its side length? Student's answer: 16

Find the distance between two vertices.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
16 is the area, not the side length. The side length is 4 units.	**4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has vertices at (0, 0), (6, 0), and (0, 8). What is its area? Student's answer: 14

Find the base and height, then use the area formula.

What did the student do wrong? Why do you think the student made this error?

The student added the base and height.
Area = 1/2 × 6 × 8 = 24 square units.

Solve the problem correctly. Show all of your work.

24 square units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle has vertices at (1, 1), (5, 1), (5, 3), and (1, 3). What is its perimeter? Student's answer: 8

Find the side lengths, then add them.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found the area of 4 × 2 = 8. **The perimeter is 4 + 2 + 4 + 2 = 12 units.**	**12 units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the length of the horizontal segment from (-2, 3) to (5, 3)? Student's answer: 3

Find the distance along the horizontal line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only from 0 to 3. The distance is 5 - (-2) = 7 units.	**7 units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the length of the vertical segment from (2, -3) to (2, 4)? Student's answer: -7

Find the distance along the vertical line.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Length is never negative. The distance is 4 - (-3) = 7 units.	**7 units**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A polygon has vertices at (0, 0), (2, 0), (2, 3), and (0, 3). What is its area? Student's answer: 10

Use the rectangle area formula.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
10 is the perimeter. The area is 2 × 3 = 6 square units.	**6 square units**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the surface area of a cube with edges of 3 cm. Student's answer: 36 square cm

Find the area of each face, then add all six.

What did the student do wrong? Why do you think the student made this error?

Each face is 3 × 3 = 9 square cm. Six faces make 6 × 9 = 54 square cm.

Solve the problem correctly. Show all of your work.

54 square cm

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the surface area of a box that is 2 by 3 by 4 units. Student's answer: 26

Find the area of each pair of faces, then add.

What did the student do wrong? Why do you think the student made this error?

The student added the dimensions. The areas are 2(6) + 2(8) + 2(12) = 52 square units.

Solve the problem correctly. Show all of your work.

52 square units

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the surface area of a box that is 5 by 2 by 1 inches. Student's answer: 17

Find the area of each pair of faces, then add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot to count each pair of faces twice. The total is 2(10 + 2 + 5) = 34 square inches.	**34 square inches**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the surface area of a cube with edges of 1/2 foot. Student's answer: 3

Find the area of one face, then multiply by 6.

What did the student do wrong? Why do you think the student made this error?

Each face is 1/2 × 1/2 = 1/4 square foot. Six faces make 6 × 1/4 = 3/2 square feet, which is 1.5.

Solve the problem correctly. Show all of your work.

1.5 square feet

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the surface area of a box that is 4 by 4 by 2 inches. Student's answer: 32

Find the area of each pair of faces, then add.

What did the student do wrong? Why do you think the student made this error?

The student counted only one of each pair of faces. The total is 2(16 + 8 + 8) = 64 square inches.

Solve the problem correctly. Show all of your work.

64 square inches

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the surface area of a cube with edges of 4 cm. Student's answer: 64

Find the area of one face, then multiply by 6.

What did the student do wrong? Why do you think the student made this error?

The student found the area of one face times 4. Each face is 16 square cm, and six faces make 96 square cm.

Solve the problem correctly. Show all of your work.

96 square cm

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 'How many pets does each student in our class have?' a statistical question? Student says: No

Decide if the answers vary.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Students will give different answers, so the question has variability. It is a statistical question.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 'What color is the teacher's shirt today?' a statistical question? Student says: Yes

Decide if one answer is expected.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There is one answer for the teacher's shirt, so there is no variability. It is not a statistical question.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 'How long does it take students to walk to school?' a statistical question? Student says: No

Decide if the answers vary.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Walking times vary from student to student, so it is a statistical question.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 'What is 5 × 7?' a statistical question? Student says: Yes

Decide if one answer is expected.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The answer is always 35, so there is no variability. It is not a statistical question.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 'What is the highest temperature in Ohio today?' a statistical question? Student says: Yes

Decide if one answer is expected.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The question has one answer for the day, not a set of varying answers. It is not a statistical question.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 'How old is the principal?' a statistical question? Student says: Yes

Decide if one answer is expected.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The question asks about one person, so there is one answer. It is not a statistical question.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the range of the data: 2, 4, 4, 6, 9. Student's answer: 9

Subtract the smallest value from the largest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
9 is the largest value, not the range. The range is 9 - 2 = 7.	**7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the median of 3, 5, 5, 7, 8, 10. Student's answer: 5

Find the middle value or values.

What did the student do wrong? Why do you think the student made this error?

With six values, the median is the mean of the two middle numbers, 5 and 7. (5 + 7) ÷ 2 = 6.

Solve the problem correctly. Show all of your work.

6

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the mean of 10, 12, and 14. Student's answer: 10

Add the values and divide by the number of values.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
10 is the smallest value, not the mean. The mean is (10 + 12 + 14) ÷ 3 = 12.	**12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which measure is affected by an outlier: the mean or the median? Student's answer: median

Think about how an outlier changes the total.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
An outlier changes the total, so the mean is affected. The median is the middle value and is less affected.	**Mean**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the range of 5, 7, 9, 11. Student's answer: 4

Subtract the smallest value from the largest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted the number of values. The range is 11 - 5 = 6.	**6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

bution has most of its data on the right and a tail to the left. Which way is it skewed? Student's answer: righ

Look at the direction of the tail.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The tail points left, so the distribution is skewed left.	**Left skew**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the mean of 1, 2, 3, 4, and 15. Student's answer: 3

Add all values and divide by 5.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 is the median, not the mean. The mean is (1 + 2 + 3 + 4 + 15) ÷ 5 = 25 ÷ 5 = 5.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A set of 5 test scores has a mean of 7. Student's answer: The sum is 2

Multiply the mean by the number of values.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided instead of multiplying. Sum = 7 × 5 = 35.	**35**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Four numbers have a mean of 8. What is their sum? Student's answer: 2

Multiply the mean by the number of values.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided 8 by 4. The sum is 8 × 4 = 32.	**32**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the median of 12, 3, 8, 20, and 5. Student's answer: 12

Order the values, then find the middle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student chose a value that is not in the middle. Ordered: 3, 5, 8, 12, 20. The median is 8.	**8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the mode of 2, 3, 3, 5, 7, 3. Student's answer: 2

Find the value that appears most often.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student picked the first value. The number 3 appears three times, so the mode is 3.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the mean of 10, 14, and 18. Student's answer: 13

Add the values and divide by 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sum is 42. Divide by 3 to get 14.	**14**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ot has 3 dots above the value 2 and 4 dots above the value 3. How many data values are there? Student's a

Count every dot.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only the dots above 3. Each dot is one value, so there are 3 + 4 = 7 values.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In a box plot, what does the line inside the box show? Student's answer: the mean

Read the parts of a box plot.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The line inside the box is the median, which splits the data in half.	**The median**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In a dot plot, what does each dot represent? Student's answer: the mean

Think about what a dot shows.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each dot represents one data value. It does not show the mean.	**One data value**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

-and-leaf plot has stem 4 with leaves 1, 5, and 8. How many data values are in the stem 4 row? Student's ar

Count the leaves.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted the stem. There are 3 leaves, so there are 3 data values: 41, 45, and 48.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box plot's lowest value is 10 and its highest value is 30. What is the range? Student's answer: 30

Subtract the minimum from the maximum.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
30 is the maximum, not the range. The range is 30 - 10 = 20.	**20**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A stem-and-leaf plot has stem 3 with leaves 0, 2, and 2. What are the values? Student's answer: 30, 32

Combine each stem with each leaf.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student left out one value. The values are 30, 32, and 32.	**30, 32, 32**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

student says: For the data 8, 8, 8, the average is 8 and the spread is 0. Is the student correct? Student says:

Check the center and spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
All values are 8, so the center is 8 and there is no spread. The summary is correct.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Test scores are 70, 75, 80, 85, and 90. The median is 80. What is the range? Student's answer: 15

Subtract the smallest from the largest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 75 from 90. The range is 90 - 70 = 20.	**20**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

as values mostly between 40 and 60, with one value at 100. Which best describes the shape? Student says

Look at where the data is clustered and where the outlier is.

What did the student do wrong? Why do you think the student made this error?

The outlier at 100 creates a tail on the right, so the shape is skewed right, not symmetric.

Solve the problem correctly. Show all of your work.

Skewed right

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The data is 2, 2, 3, and 9. Which measure of center is the best to report? Student's answer: mean

Think about the outlier.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The outlier 9 pulls the mean up. The median is a better summary of the typical value here.	**Median**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A data set has 11 values. Which position holds the median? Student's answer: 5th

Find the middle position.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The middle position of 11 values is (11 + 1) ÷ 2 = 6, so the median is the 6th value.	**6th**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Data goes from 1 to 9 with a mean and median of 5. What is the range? Student's answer: 9

Subtract the smallest value from the largest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
9 is the largest value. The range is 9 - 1 = 8.	**8**